



Critical heat flux prediction through machine learning model for narrow rectangular channels

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ABSTRACT

Accurately predicting critical heat flux (CHF) is crucial for ensuring efficient energy transfer and safe operation in thermal system, particularly in narrow rectangular channels, which are favored for their superior thermo-hydraulic performance and compactness. This study compiles 660 experimental CHF data points from narrow rectangular channels under wide variety of operating conditions and evaluates existing CHF correlations. Four machine learning (ML) models—back-propagation (BP) neural networks, random forest (RF), support vector regression (SVR), and Long Short-Term Memory (LSTM) are employed to predict CHF using input parameters encompassing thermal-hydraulic and geometrical characteristics. The results indicate that the BP approach has a high level of performance, with a root-mean-square error (RMSE) of 38.37 and a mean absolute error (MAE) of 24.33. To further optimize the BP neural network, four metaheuristic algorithms—particle swarm optimization (PSO), genetic algorithms (GA), cuckoo search (CS), and ant colony algorithm (ACA) are applied. The GA optimization yields the most accurate predictions, achieving a RMSE of 33.53 and a MAE of 19.13. The trained GA-BP model exhibits robust performance in predicting CHF in narrow rectangular channels under varying pressure conditions.

1. Introduction

Plate-type fuels are highly favored for their compact structure and high volumetric heat release rate, making them a prevalent choice in experimental research reactors. Within these assemblies, fuel plates are arranged to form narrow rectangular channels. During reactor operation, coolant enters one end of the module and exits the other, effectively removing the fission heat produced at the fuel plate surfaces through convective heat transfer [1]. Compared to conventional circular channels, rectangular channels are arranged in a more condensed manner and, as a result of the distribution of surface tension between the channel walls and internal fluid, demonstrate increased friction pressure drop and enhanced heat transfer capabilities [2]. These characteristics make them particularly well-suited for the miniaturization of nuclear power units, enabling effective dissipation of high heat flux.

The working fluid flows and boils within a closed, narrow rectangular heating channel, driven by external forces, as illustrated in Fig. 1 (a). As shown in Fig. 1(b), the fluid enters from the left boundary and exits at right boundary. Bubbles nucleate at the channel wall, aggregate,

and eventually detach from the fluid along the flow direction. The location where the wall dries out is identified as the boiling crisis point, also referred to as the Dryout type critical heat flux (CHF) [3]. The bulk liquid associated with a Dryout type boiling crisis is primarily in a saturated state. Continued heating leads to the complete evaporation of dispersed droplets, resulting in a transition to a single-phase vapor flow. CHF is an important phenomenon that significantly affects the economic efficiency and safety of equipment, characterized by a sharp decline in the heat transfer coefficient and a sudden rise in wall superheat.

The experiment is crucial yet challenging for obtaining CHF data. Wang et al. [4] conducted experimental research on CHF for different working fluids. Their results demonstrated that the heat transfer coefficient was 60.73 % higher than that of reverse osmosis water at CHF. As the heating angle increased, CHF also increased, albeit at a diminishing rate. To explore the mechanism underlying the enhanced heat transfer performance of nanofluids, post-experimental analyses, including scanning electron microscopy, contact angle measurements, and surface roughness characterizations, were performed on the heating surface. Yan et al. [5] visually observed the flow of deionized water for a narrow

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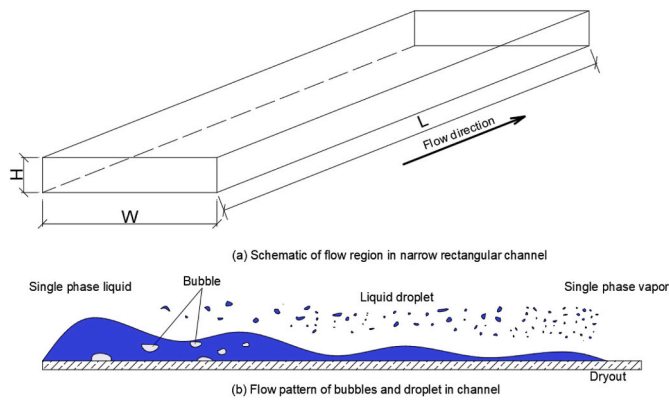


Fig. 1. The schematic of narrow rectangular channel and flow pattern.

rectangular channel. The CHF correlations proposed by Sudo et al. [6] exhibited the best predictive performance among the applied correlations. Nevertheless, significant uncertainty remained in the Sudo correlation when compared to experimental data. Visualization images from the channels provided robust support for analyzing the mechanism of boiling crisis occurrence. Kim et al. [7] experimentally investigated CHF in a narrow rectangular channel with two-sided heating under upward flow conditions. The results revealed significant discrepancies between the correlation predictions and the experimental data, and the study was confined to low-pressure conditions. The evaluation of engineering designs aimed at promoting or preventing CHF continues to rely on costly experiments. Furthermore, the experimental loop test range is restricted, particularly for high-pressure conditions, making safety and stability difficult to ensure.

The numerical simulation method appears to be a viable option. Some researchers have employed numerical models to study CHF, applying the narrow rectangular channel structures. Meehen et al. [8] developed a computational model based on RELAP5-3D, enhancing the existing CHF model to mechanistically account for transient effects, material properties, and RISA effects by comparing it with experimental CHF data. However, they found that transient CHF and nucleate boiling heat transfer could not be predicted precisely. There 1-D model in RELAP5-3D had limitations in incorporating incorporate physical surface attributes. The development and implementation of novel correlations were essential to resolve the discrepancies between simulations and experiments. Han et al. [9] proposed a CHF model for narrow rectangular channel using a two-fluid Eulerian CFD framework. Their predictions were validated against 22 experimental data points spanning diverse operational conditions, incorporating a focused examination of phase change dynamics within narrow channels. This analysis encompassed a broad spectrum of study parameters and facilitated a systematic evaluation of model accuracy. Certain models demonstrated superior predictive performance. The numerical approach effectively simulated flow and heat transfer processes, illustrating reasonable accuracy; however, it demands substantial calculation resource, and the model's underlying mechanisms require refinement through systematic experimental validation.

Empirical correlations can therefore be established through regression analysis of experimental data. These correlations are commonly employed in thermal-hydraulic design due to their ease of implementation and practical applicability in engineering contexts. Numerous CHF correlations exist for narrow rectangular channels, however, these require refinement and evaluation to ensure universality and applicability [10]. Alternatively, Look-Up Tables offer a cost-effective method for CHF determination across a broad range of thermodynamic circumstances, utilizing parameters such as pressure, mass flow, and quality. Nevertheless, existing studies indicate that the Groeneveld Look-Up Table may overpredict CHF values in narrow rectangular channels, particularly under low-pressure and low-mass-flux conditions

[11].

Machine learning (ML) methodology is a data-driven statistical approach that prioritizes empirical data over physics-based modeling. While ML algorithms are primarily influenced by training datasets and do not require physics-based foundations, their underlying assumptions must align with the dataset's scope to ensure reliability. Recent advancements have expanded ML applications to thermal fluid sciences [12] and mechanical systems [13], demonstrating promising results in CHF prediction. Kumar et al. [14] evaluated three models for artificial neural network (ANN), random forest (RF) and support vector machine (SVM). They discovered that the ANN model exhibited a significantly lower error rate, indicating a strong correlation with the experimental data. Despite covering a wide range of operating conditions, the experimental dataset of Becker et al. [15] was still regarded relatively limited in size. Niu et al. [16] combined deep feed-forward neural networks in conjunction with empirical correlations to predict both CHF magnitudes and their spatial locations reducing prediction errors from 10 % to 5 % compared to conventional neural network methodologies. Kim et al. [17] developed a ML model trained on experiment and pseudo data generated by CHF empirical correlations, achieving the closest agreement with the experimental results (average absolute error: 0.9 %). Yan et al. [18] proposed a BP neural network for the prediction of CHF. The prediction errors of BP neural networks exhibited a maximum deviation of 20 %. Regarding BP-based approaches, Li et al. [19] quantified the sensitivity coefficients of input parameters through partial differential equation analysis. Hedayat [20] investigated the learning capacity of the model output by examining varying numbers of hidden layers. Khalid et al. [21] introduced a predictive model integrating an autoencoder and a meta-learner, which was constructed on top of an ensemble of deep sparse autoencoders. This unsupervised learning approach promotes diversity in decision-making and facilitates a more comprehensive exploration of the solution space, resulting in higher accuracy compared to individual models. The proposed method demonstrates improved accuracy in predicting CHF. Furthermore, the authors extended their analysis beyond conventional input variables by incorporating geometric and flow parameters, fluid physical properties, and dimensionless groups in their machine learning framework [22]. Among the four ML models evaluated in their study, the ANN model exhibited superior performance, as indicated by its lower MAE of 8.85 %. A.S. Nafey [23] developed a 7-9-1 ANN architecture to predict CHF in circular pipes. The experimental CHF data were subsequently compared with predictions from both the proposed ANN-based correlation and an existing best-fit correlation. The study revealed maximum deviations of 5.5 % and 30 % for the ANN model and conventional correlation, respectively.

It's worth noting that the available data on CHF in most existing research is limited, especially for the experimental data at high pressures. This limitation poses a challenge for building and training ML algorithms. Furthermore, while certain algorithms may be better suited for current prediction tasks due to advantages such as accuracy, robustness, and stability, the diversity of optimized algorithms remains relatively restricted. This narrow scope may hinder the ability to fully demonstrate the superiority for proposed method. In present research, CHF experimental data in narrow rectangular channels has been summarized across different operational ranges in recent years. These data have been properly combined to derive the optimal correlation based on the data collection methodology. Additionally, the optimal combination of various ML algorithms has been identified for the limited CHF dataset in narrow rectangular channels.

The aim of this study is to develop an accurate model for predicting CHF in narrow rectangular channel across a wide variety of pressure conditions. Prior to modeling, a dataset consisting of 660 samples obtained from peer-reviewed literature is established. These data are also used to evaluate several classic CHF correlations. The selection of input parameters for the ML model is examined and optimized based on the correlations of the CHF data. Four distinct ML models are utilized to

predict CHF, with the most suitable model selected based on prediction performance. In addition, another four different ML algorithms are employed for optimize the selected model. The complete research methodology for CHF prediction is illustrated in Fig. 2. The objective of this study is to develop a systematic approach for predicting CHF for narrow rectangular channels.

2. CHF data analysis and evaluation

In this section, we collect CHF data from recent years and evaluate several factors for narrow rectangular channels. The dataset covers a wide range of operational conditions. In addition, some basic CHF correlations are summarized and assessed based on the collected data.

2.1. CHF data collection

Data collection focuses on narrow rectangular channels across a broad range of parameters. Since this study is limited to narrow rectangular channels, data covering a wide pressure range are included. However, high-pressure data lacking complete thermal-hydraulic and geometric parameters are excluded. Table 1 presents the CHF data collection from the literature, comprising a total of 660 data points. The experimental dataset encompasses a wide range of conditions for water as the working fluid in narrow rectangular channels.

Violin plots are employed to visualize the full distribution of each feature using the kernel density estimation (KDE) technique with a Gaussian kernel, as shown in Fig. 3. These plots offer a statistical representation of data ranges, as outlined in Table 1, and highlight frequently observed values in the dataset. The distribution of data is shown in Fig. 4. Furthermore, violin plots are particularly useful for analyzing bimodal distributions.

The impact of input features on output parameters significantly influences the accuracy of ML algorithms. This study evaluates the impact of input attributes by employing three correlation coefficient (R) approaches: Pearson's, Spearman's, and Kendall's. The range of $\cdot R$ spans from -1 to $+1$. A value of -1 represents a fully negative correlation, 0 indicates no correlation, and $+1$ denotes a perfect positive correlation. The correlation coefficients are defined by Eqs. (1)–(3) according to specific requirements [28].

$$\rho_{\text{person}} = \frac{\sum_{i=1}^{n_s} (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n_s} (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^{n_s} (y_i - \bar{y})^2}} \quad (1)$$

$$\rho_{\text{sperman}} = \rho_{\text{person}} \frac{\text{cov}(x, y)}{\gamma_x \gamma_y} \quad (2)$$

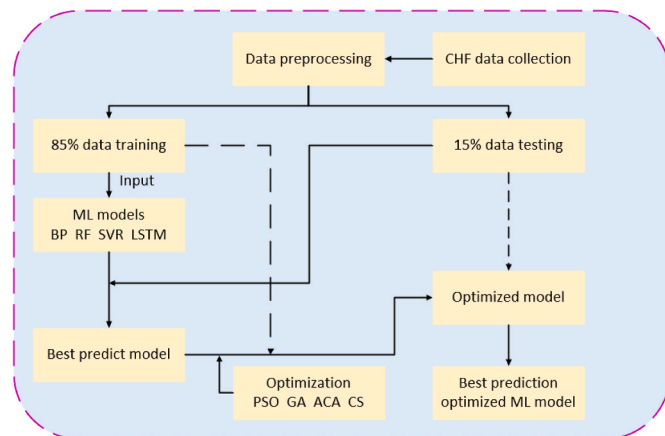


Fig. 2. The whole research process for prediction of the CHF for narrow rectangular channel.

$$\tau_{\text{kendall}} = \frac{2(n_c - n_d)}{k(k-1)} \quad (3)$$

where n_s denotes the total number of samples, $\text{cov}(x, y)$ represents the covariance between x and y variables, γ_x and γ_y indicate standard deviations, and n_c and n_d denote the number of concordant and discordant pairs, respectively. Furthermore, a heatmap is utilized to illustrate the magnitude of correlation coefficients between input parameters. The heatmap is a visualization tool that employs a variety of colors to represent data relationships within a tabular format. Collinearity issues may arise when input parameters exhibit a strong correlation. In order to circumvent this issue, linearly independent input parameters are implemented.

The selection of an appropriate number of input parameters is crucial for accurate predictions. Wang et al. [29] found that the overall deviation decreased from 59.75 % with a single parameter to a minimum of 2.63 % when using five parameters. Additionally, they suggested that using inlet parameters instead of derived outlet parameters improves prediction accuracy. Consequently, hydraulic parameters (mass flux, subcooling, and pressure) and geometric parameters (wetted width, heated length, and gap size) are initially selected to ensure sufficient characterization and evaluate correlations for different parameters. The correlations for 6 input parameters are shown in Fig. S1. The geometric parameter H exhibited low relevance to q_{CHF} , necessitating its integration to improve its relevance.

The hydraulic equivalent diameter (D_{hed}) converts non-circular geometries into an equivalent "circular" dimension, enabling the application of well-established correlations for circular pipes. The equivalent diameter inherently assigns greater weight to gap size due to its inverse relationship with wetted perimeter, aligning with the physical significance of this dimension. Additionally, it integrates both height and width, avoiding oversimplification. Most CHF correlations for non-circular channels are experimentally derived using D_{hed} as a key variable. Therefore, D_{hed} is considered one of the geometric parameters and used to evaluate the correlations. As shown in Fig. S2, the correlations for 5 input parameters are illustrated. The relevance between D_{hed} and q_{CHF} shows slight improvement but remains insignificant, indicating the need for further enhancement.

The Length-to-Hydraulic-Equivalent-Diameter ratio (L/D_{hed}) encompasses the heated length and represents the flow development process. In experiments, this ratio is used in scaling studies where flow development time and space are significant. Additionally, in predicting CHF in narrow rectangular channels, L/D_{hed} as an input parameter demonstrates high prediction accuracy with a limited dataset [18]. Therefore, L/D_{hed} is ultimately regarded as a key geometric parameter. The correlation for the 4 input parameters is shown in Fig. 5, where the relationship between L/D_{hed} and q_{CHF} is notably improved. As a result, both the geometric parameter (L/D_{hed}) and the hydraulic parameters (mass flux, inlet subcooling, and pressure) are selected as input parameters for ML algorithms in CHF prediction.

2.2. Previous CHF correlations evaluation

Empirical correlations are derived from the extant experimental data through regression analysis. They are straightforward to implement and provide reliable accuracy within their application range, making them extensively employed in thermal-hydraulic design from an engineering perspective.

Several CHF empirical correlations for rectangular channel are summarized and discussed. Table 2 shows the CHF correlations along with their applicable ranges. These parameters for dimensionless CHF, mass flux and inlet subcooling can be described as

$$q_{\text{CHF}}^* = \frac{q_{\text{CHF}}}{h_{\text{fg}} \sqrt{\lambda(\rho_f - \rho_g) \rho_g g}} \quad (4)$$

Table 1
CHF data collection based on the literature.

Source	Width, W [mm]	Height, H [mm]	Hydraulic diameter, D_{hed} [mm]	Heated length, L [mm]	Mass flux, G [kg/ m ² s]	Inlet subcooling, ΔT_{in} [K]	Pressure, P [MPa]	data number
Du et al. [24]	20, 60	1.1, 2.5	2.09-4.44	150–690	30–4300	0–301	1–14	296
Huang et al. [25]	60	0.5–3	0.99-5.71	750	50–200	10–60	1–8	312
Tang et al. [26]	50	2	3.84	1000	300–1000	20–140	12–15	22
He et al. [27]	70	2.7	5.20	700	135–711	20–70	0.14	30
Total	20–70	0.5–3	0.99-5.71	150–1000	30–4300	0–301	0.14–15	660

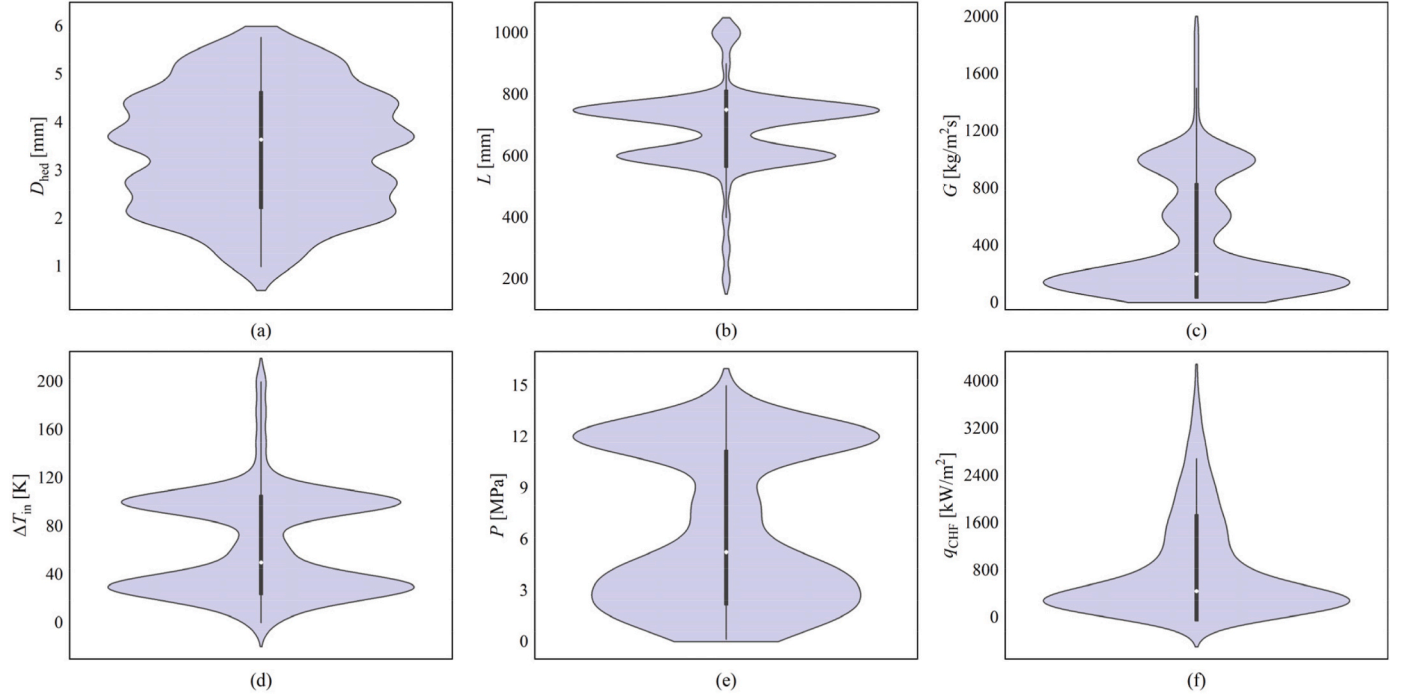


Fig. 3. The KDE distributions of the parameters illustrated through violin plots.

$$G^* = \frac{G}{\sqrt{\lambda(\rho_f - \rho_g)\rho_g g}} \quad (5)$$

$$T_{sub}^* = \frac{c_p \cdot \Delta T_{sub}}{h_{fg}} \quad (6)$$

where λ is critical wavelength and calculated by

$$\lambda = \left[\frac{\sigma}{(\rho_f - \rho_g)g} \right]^{0.5} \quad (7)$$

Fig. 6 compares the measured CHF values with predictions from various correlations. The CHF data falls within the applicable range of these correlations. As shown in Fig. 6(b), the Zhou correlation underestimates CHF across all experimental data points. In contrast, the correlations proposed by Debortoli and Sudo provides more accurate predictions. Nevertheless, certain deviations persist, necessitating further evaluation of prediction performance using additional statistical metrics.

Quantitative assessment is provided by using the coefficient of determination (R^2), mean absolute error (MAE), mean absolute percentage error (MPAE) and root-mean-square error (RMSE), with the following formulas for the four statistical indicators [33].

$$R^2 = 1 - \frac{\sum_{i=1}^N (y_i - y_i^*)^2}{\sum_{i=1}^N (y_i - \bar{y})^2} \quad (8)$$

$$MAE = \frac{1}{N} \sum_{i=1}^N |y_i - y_i^*| \quad (9)$$

$$MPAE = \frac{1}{N} \sum_{i=1}^N \left| \frac{y_i^* - y_i}{y_i} \right| \quad (10)$$

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i^* - y_i)^2} \quad (11)$$

where the term of N represents the number of sampled values, $\bar{y}$ is the average of all actual values, y_i^* and y_i correspond to the predicted and experimental CHF data, separately.

Table 3 presents a comparative analysis of the quantitative assessment results. Among the correlations, only Debortoli exhibits a positive R^2 value, indicating predictive capability. In contrast, the other correlations perform poorly, displaying negative R^2 value. This discrepancy arises due to weak variable correlations and potential errors in the experimental data. Furthermore, the MAE values for all correlations are

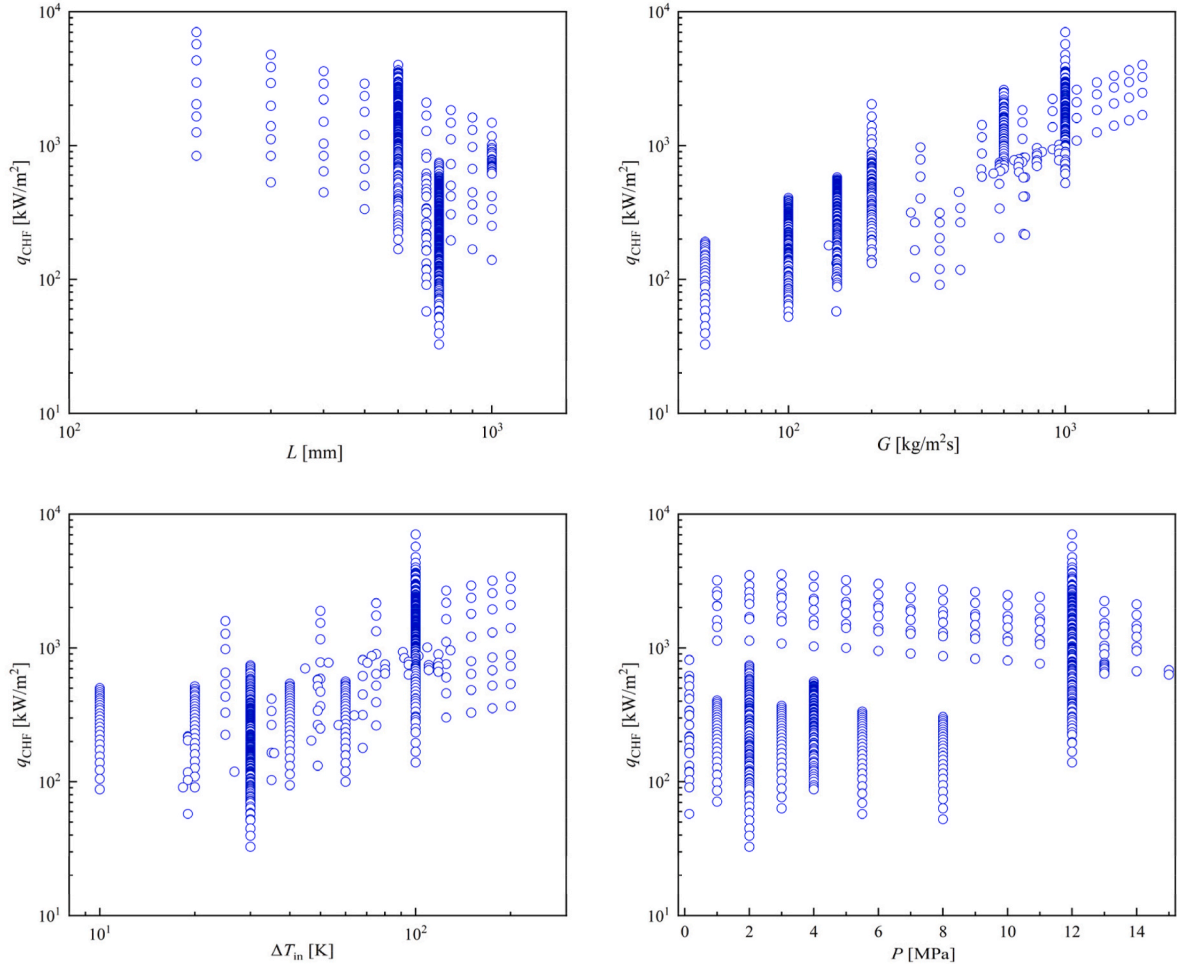


Fig. 4. The distribution of CHF data across different input parameters.

relatively high. Although MPAE and RSME remain below 0.4, the overall error is still substantial.

Given the low prediction accuracy, these correlations are not suitable for engineering applications. The CHF data presented in this article pertain to narrow rectangular channels under various experimental conditions. The reliance on empirical CHF correlations inevitably introduces significant errors when prediction CHF in narrow rectangular channels. Therefore, a more appropriate approach is necessary for accurately predicting CHF data under experimental conditions.

3. ML introduction

The current work utilizes four distinct methods, including BP, support vector regression (SVR), RF, and long short-term memory (LSTM), to predict CHF. In recent CHF prediction studies, these ML models are commonly used and demonstrate advantages in prediction [34]. Therefore, they are selected for CHF prediction for narrow rectangular channels. Furthermore, based on the prediction results, four additional optimization algorithms are incorporated: particle swarm optimization (PSO), genetic algorithms (GA), cuckoo search (CS), and ant colony algorithm (ACA). This section introduces an overview of these algorithms and explains their computational procedures. In present research, the input variables are L/D_{hed} , G , ΔT_{in} and P , while q_{CHF} is the output variable.

3.1. ML methods

3.1.1. BP neural network

The BP neural network is a multilayer feedforward neural network trained using the error backpropagation algorithm. It demonstrates strong adaptability and exceptional multidimensional nonlinear mapping capabilities, rendering it one of the most extensively employed ML techniques [35]. The topological structure of a BP method is illustrated in Fig. 7. This network is composed of an input layer, one or more hidden layers, and an output layer. The input layer receives data, which is then processed and subjected to computations involving weights and biases, ultimately generating output values. These output values are compared with expected values to calculate training errors. Using error backpropagation, the network iteratively minimizes the sum of squared errors between actual and predicted outputs by adjusting the weights and thresholds of each layer.

The term for ω_{ij} and ω_{jk} are the weight for middle layer and output layer respectively in Fig. 7. The amount of nodes in the hidden layer is normally given as shown in Eq. (12) [36].

$$h = \sqrt{m + n} + a \quad (12)$$

where m and n are the number of input layer and output layer nodes respectively, a is the adjustment constant of 1–10, h is the number of hidden layer nodes. Many scholars have proposed empirical equations or determined optimal hidden layer node numbers through experimentation to guide node selection. However, these approaches still present certain limitations. To address this, we systematically teste

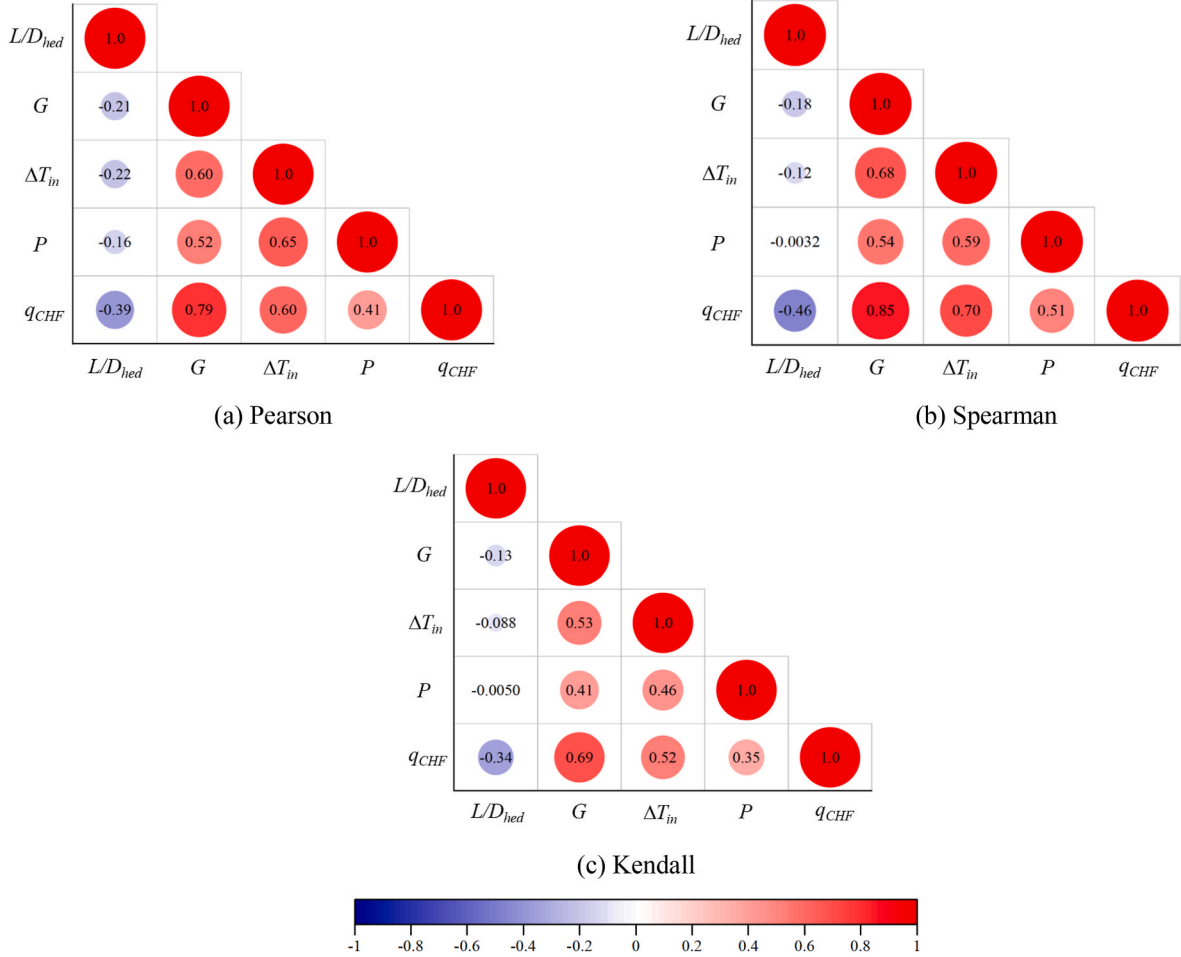


Fig. 5. Heat map correlation for 4 input parameters.

Table 2
Applicable CHF correlations for rectangular channels.

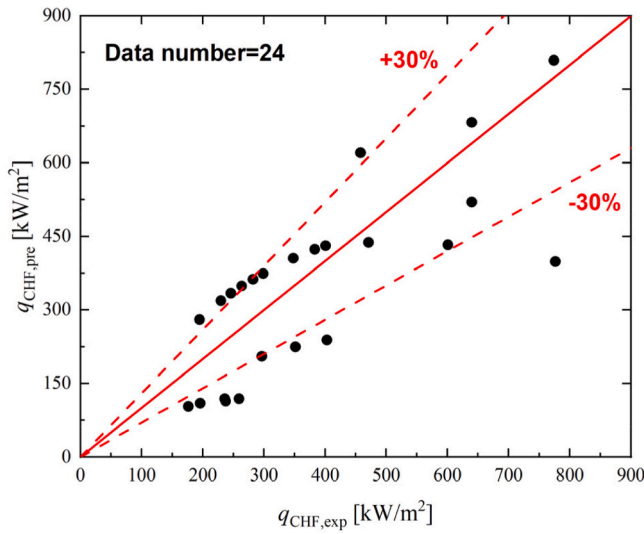
References	CHF correlations	Remarks
Debortoli et al. [30]	$2152 \text{ kg/m}^2\text{s} < G < 6725 \text{ kg/m}^2\text{s}$ $q_{CHF} = 1.36 \times 10^4 \left(\frac{2.326 H_{BO}}{10^3} \right)^{2.5} \left(1 + \frac{1.35 G}{10^{10}} \right) e^{-0.0012 L/H}$ $269 \text{ kg/m}^2\text{s} < G < 2152 \text{ kg/m}^2\text{s}$ $q_{CHF} = 1.36 \times 10^4 \left(\frac{2.326 H_{BO}}{10^3} \right)^{2.5} e^{-0.0012 L/H}$	P : 4–13.8 MPa, ΔT_{sub} : 8–540 K, G : 40–6431 kg/m ² s, D_{hed} : 2.83–4.91 mm L : 152.4–685.8 mm
Zhou et al. [31]	$q_{CHF} = \frac{(C \cdot h_{fg} + \Delta h_m) G}{128.65 G^{0.09143} + \frac{4L}{D_{hed}}} \left\{ 1 + \frac{7.8379}{P} \exp \left(-\frac{[\ln(P/7.0326)]^2}{0.1871} \right) \right\}$ where, $C = \frac{1.83}{1 + 0.0308 G^{0.9032} (1000 D_{hed})^{-1.3423}}$	P : 1.9–15.8 MPa ΔT_{sub} : 41–280 K G : 293–2534 kg/m ² s L : 200–1200 mm
Sudo et al. [32]	$q_{CHF}^* = 0.005 \cdot G^* ^{0.611} \cdot \frac{1 + \frac{5000}{ G^* } \cdot \Delta T_{sub}^*}{1 + 25 \cdot G^* ^{-1.389} \frac{A_H}{A_L}}$ $G^* > G_2^*$ $q_{CHF}^* = 0.7 \frac{A_L}{A_H} \frac{\sqrt{W/\lambda}}{[1 + (\rho_g/\rho_f)^{1/4}]^2} \cdot G^* < G_2^*$ $G_2^* = \left\{ 140 \frac{A_L}{A_H} \frac{\sqrt{W/\lambda}}{[1 + (\rho_g/\rho_f)^{1/4}]^2} \right\}$	P : 0.1–4 MPa G : 0–6250 kg/m ² s ΔT_{sub} : 1–213 K

different parameter ranges based on error minimization to identify the best-performing configuration. The optimal values are determined through this process. In the next section, we discuss on the optimal configuration of hidden layers and neurons in detail.

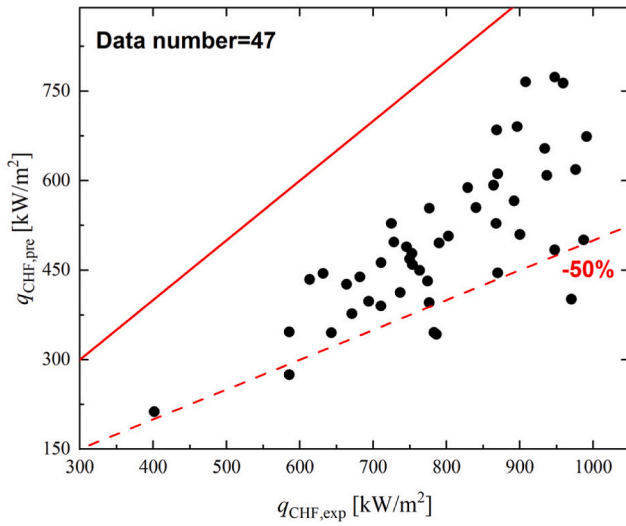
3.1.2. RF method

The RF algorithm employs multiple decision trees for data analysis in both classification and regression tasks. Every classification and regression tree (CART) or decision tree originates from a unique root node and expands through branching. Unlike traditional approaches, RF utilizes only a subset of the dataset for training purpose. Each tree division represents a feature, which is associated with a specific threshold value. Based on input data, left and right child nodes are generated depending on whether the data meets the threshold criteria.

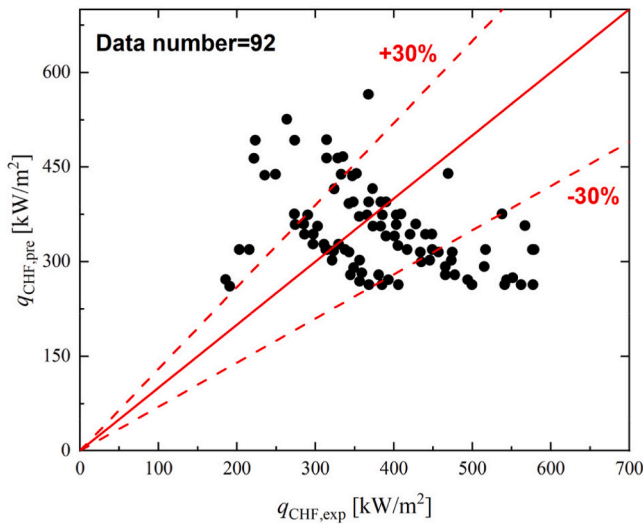
The input data is repeatedly tested and divided until a final node is reached, which generates a constant value. A random forest is an ensemble of decision trees that consists of multiple decision trees, each constructed using a distinct combination of features and data from the training set. RF is a ML algorithm that utilizes this ensemble approach, with each tree trained on a random subset of features. Each tree independently selects the most likely outcome, and the final forecast is determined by a majority vote. Fig. 8 illustrates the working principle of the RF model. Our study involved a methodical investigation of several hyperparameter values for the RF model, conducting experiments with varying parameter ranges for trees and nodes to determine the optimal values. The gini splitting criterion is employed.



(a) The Dehortoli correlation prediction



(b) The Zhou correlation prediction



(c) The Sudo correlation prediction

Fig. 6. The CHF prediction using correlations.

Table 3

Performance parameters of CHF correlations.

No.	Correlation	Data number	R^2	MAE	MPAE	RSME
1	Dehortoli	24	0.50	103.86	0.30	125.34
2	Zhou	47	-5.00	295.09	0.37	308.15
3	Sudo	92	-1.20	113.07	0.30	140.28

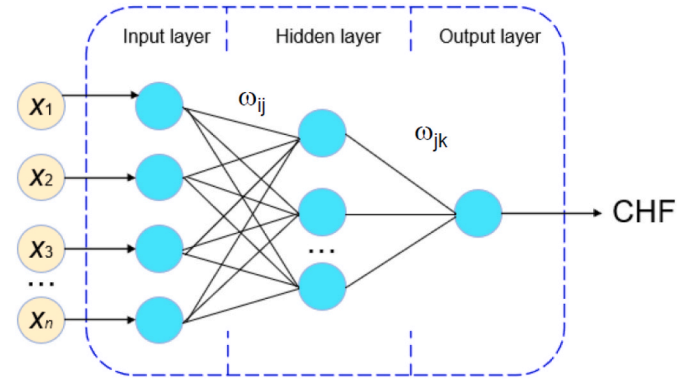


Fig. 7. The structure of established BP neural network.

3.1.3. SVR method

SVR has been derived from statistical learning theory and is designed to achieve structural risk minimization, enhancing its generalization capabilities for future data.

In regression analysis, SVR exhibits three key characteristics. Initially, SVR functioned as a linear classifier in high-dimensional spaces. Furthermore, SVR performs regression estimation by optimizing model parameters to minimize the discrepancy between predicted values and actual results. The risk function of SVR incorporates both regularization factors and empirical errors. By employing insensitive loss functions, SVR can leverage historical data to forecast future values. For a comprehensive analysis of the selection and implementation of hyperparameters in SVR, please refer to the mentioned reference [37].

3.1.4. LSTM network cell structure

LSTM is an algorithm derived from recurrent neural network (RNN) that shares a similar control flow with basic RNN. However, LSTM is far more complex than traditional RNNs and is capable of addressing the issues of vanishing and exploding gradients. LSTM exhibits superior performance in comparison to other neural network models. Examples of such models include the time-based RNNs and hidden Markov models. Furthermore, a larger deep neural network can be formed using LSTM. Since LSTM can be regarded as an intricate component of a nonlinear system model, it can be classified as a complicated nonlinear system element [38].

As shown in Fig. 9, an LSTM unit comprises forget gates, input gates, output gates, and memory cells, collectively forming the core architecture of LSTM neural network. The structure of the LSTM provides it with extraordinary efficacy in handling long sequential data and managing long-term dependencies. The h_{t-1} represents the output of the LSTM at the previous time step, while h_t represents the output at the current time step. Similarly, c_{t-1} represents the previous time step's storage unit state, and c_t represents the current time step's storage unit state. Lastly, x_t represents the input. Information can be sequentially propagated via the cell state. A detailed analysis of each gate in the LSTM network in Fig. 9.

- (1) Forget gate: The forgetting gate determines how much of the $t-1$ moment cell state is retained in the current cell state. The calculated size of the f_t will determine the cells The degree to

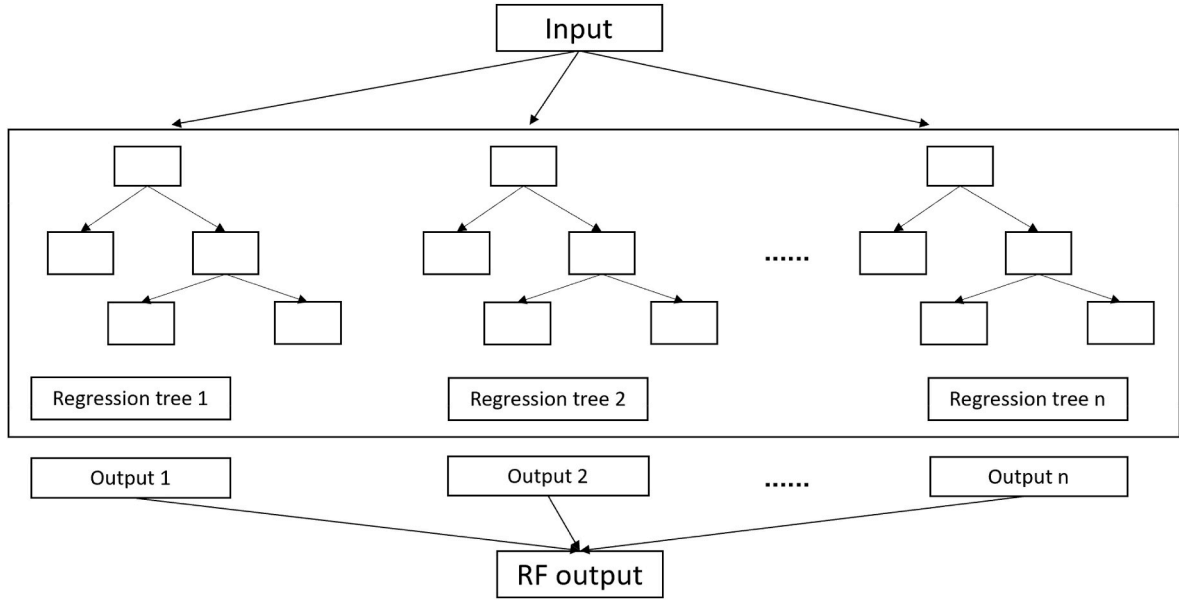


Fig. 8. Working principle of RF algorithm.

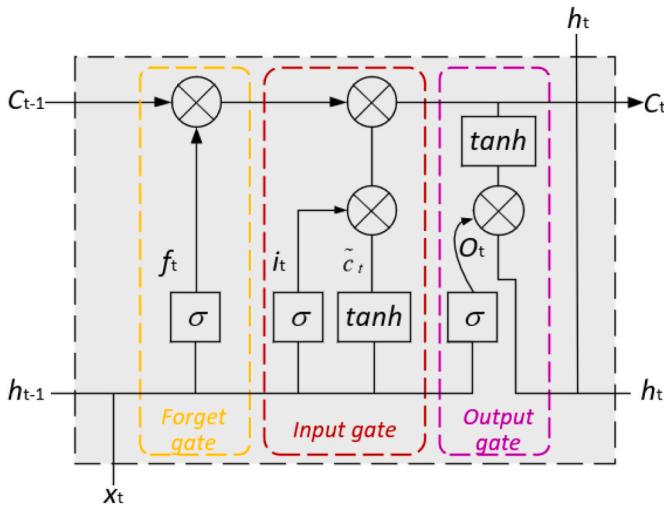


Fig. 9. The scheme of LSTM.

which state information is retained in the current unit, which can be expressed as

$$f_t = \sigma(w_f \cdot h_{t-1} + u_f \cdot x_t + a_f) \quad (13)$$

where w represents the weights associated with historical information, u corresponds to the weights for the current input variable, and a denotes the biases in the current gate.

- (2) Input gate: The input gate determines how much of the newly obtained information $\tilde{c}_t$ needs to be updated, and the computed results become part of the updated neuronal state c_t . This process can be expressed by:

$$i_t = \sigma(w_i \cdot h_{t-1} + u_i \cdot x_t + a_i) \quad (14)$$

$$\tilde{c}_t = \tanh(w_c \cdot h_{t-1} + u_c \cdot x_t + a_c) \quad (15)$$

Additionally, updating the cell state from the previous moment to the current moment is required. The cell state at the current time step is given by:

$$c_t = f_t \otimes c_{t-1} + i_t \otimes \tilde{c}_t \quad (16)$$

where c_t and c_{t-1} represent the cell states retaining current and past information, respectively.

- (3) Output gate: The final gating unit is the output gate, which functions symmetrically to the input gate and determines what information should be output. The Sigmoid activation function processes the previous cell state information h_{t-1} and the current input x_t to compute O_t :

$$O_t = \sigma(O_i \cdot h_{t-1} + O_i \cdot x_t + a_o) \quad (17)$$

The cell state information at the current time step is then transformed by the Tanh function and multiplied by O_t to obtain the output h_t :

$$h_t = O_t \cdot \tanh(c_t) \quad (18)$$

3.2. Optimized ML algorithms

3.2.1. PSO method

The PSO method is a heuristic optimization technique that draws inspiration from the collective behaviors observed in nature, such as the flocking of birds. The process involves simulating the interactions and competition among individuals in a group to identify the optimal solutions, iteratively approaching a more refined solution. PSO method is particularly well-suited for optimizing structural parameters due to its simplicity and rapid convergence, which are its key features.

The fundamental concept of the PSO algorithm is that, within a search space of a specific dimension, every member in the swarm can be seen as a search particle. The particle's search space encompasses the entire dimension space. The position of each particle within the search space represents a potential solution to the optimization problem. The updated velocity v_{id}^{k+1} and position x_{id}^{k+1} of the particle are given by:

$$v_{id}^{k+1} = \omega v_{id}^k + c_1 r_1 (p_{id,pbest}^k - x_{id}^k) + c_2 r_2 (p_{d,gbest}^k - x_{id}^k) \quad (19)$$

$$x_{id}^{k+1} = x_{id}^k + v_{id}^{k+1} \quad (20)$$

In the given formulas, the variables are defined as follows: i represents the particle number, d represents the particle dimension, k denotes the

iteration number, ω represents the inertial weight, c_1 and c_2 represent the individual and swarm learning factors, respectively, r_1 and r_2 represent random numbers within the interval $[0,1]$.

3.2.2. GA algorithm

GA is a widely employed method for evolutionary optimization. It is particularly well-suited for various optimization problems, especially those involving approximations and uncertainties in modeling. The GA was created by John Holland in 1975 using a heuristic approach. It addresses optimization issues by studying adaptability in both natural and artificial systems. The GA selects individuals from the current population to act as parents and produce offspring for the next generation at each step. The selection process is determined by the adaptability and fitness of each individual, with the goal of identifying the most suitable individual who will pass on their traits and contribute to the evolution of a successful generation. The production of the future generation from the current one is governed by two primary rules.

- Crossover rules include merging genetic information from two parents to generate offspring for the following generation;
- Mutation rules introduce random alterations to the genetic makeup of each parent to produce offspring.

The GA procedure that has been implemented is show in Fig. 10.

3.2.3. ACA method

The optimization of ACO is a population-based heuristic method, similar to the technique of Dorigo et al. PSO. This strategy is inspired by the behavior of ant colonies in their search for food. Each colony member strives to identify the most traveled path through the food source. Ants release signaling pheromones to mark their location relative to the source. The probability of path selection is determined by the saturation of the pheromone level, with ants being more likely to follow along the route with the strongest pheromone. Ants use pheromones to communicate, leaving a trace as they move. In the presence of other ants, they are capable of detecting pheromone traces and selecting a path with the higher pheromone concentration. The concentration of pheromones along the route is subsequently increased as more ants traverse it, making the route more attractive. As the number of ants selecting the path grows, the probability of subsequent ants choosing it also increases [39].

3.2.4. CS algorithm

The CS algorithm is a metaheuristic intelligent optimization technique inspired by the breeding behavior of cuckoos. It primarily utilizes the nesting parasitism of Cuckoo and Lévy flight mechanism to optimization. The algorithm is based on three idealized rules.

- Each cuckoo lays one egg at a time and places it in a randomly selected nest;
- Among the randomly selected nests, the best one will be passed on to the next generation;
- The number of available host nests is fixed, and the probability of cuckoo eggs being laid is found by the host bird. In this case, the host birds either throw away their eggs or give up their nest and build a new one.

4. Results and discussions

4.1. Comparison with other ML techniques

The CHF logs data in narrow rectangular channel including L/D_{hed} , G , ΔT_{in} , and P are used as input parameters, while the CHF value (q_{CHF}) is used as the output label for training the ML predictive models. During the training of ML models, 85 % of the dataset is used, while the remaining 15 % reserved for testing purposes. The dataset is divided into

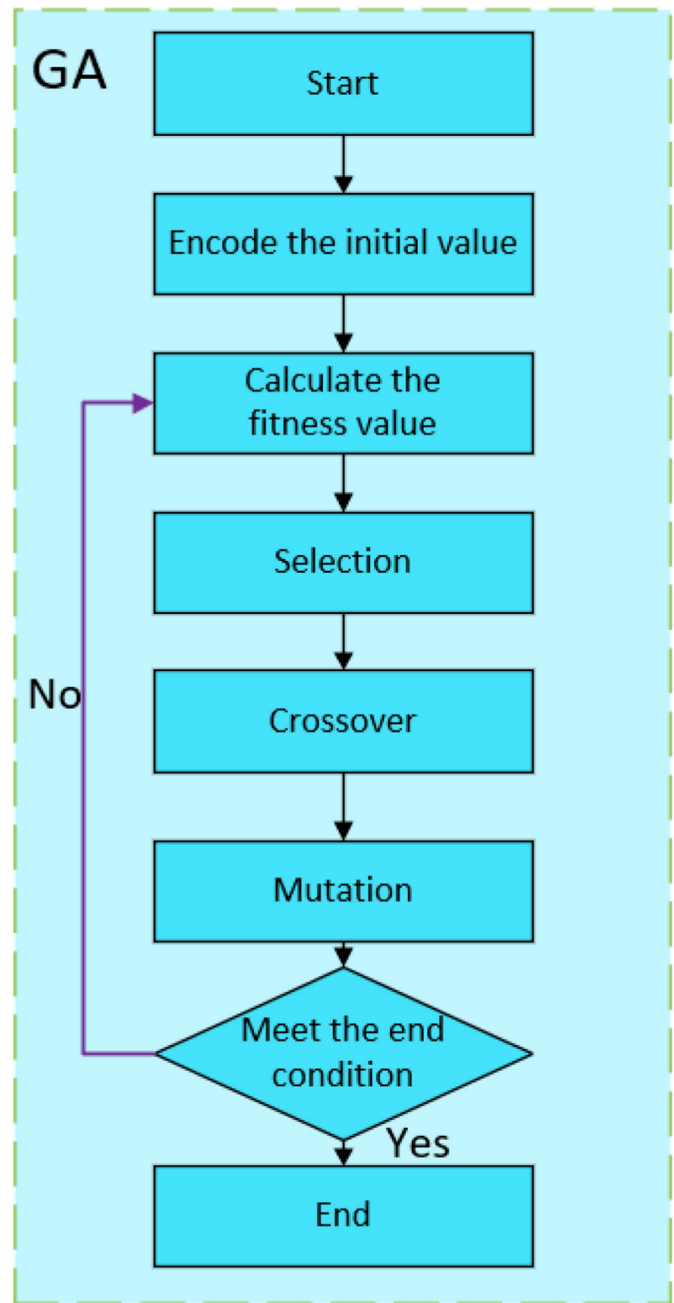


Fig. 10. The GA optimization procedure.

two subsets: 561 data points are utilized for training the model, while the remaining 91 are designated for testing purposes. We performed a sensitivity analysis on the hidden layer and the number of neurons in the BP neural network, covering a range of one to six layers and 5 to 30 neurons. Fig. 11 illustrates the sensitivity outcomes of various layer configurations and the number of neurons, corresponding with their related RMSE values. Below the shaded region is a histogram, while above it is a 3D graph. Distinct distributions and trends are clearly observed across different hidden layers and numbers of neurons. The sensitivity analysis revealed that a neural network architecture consisting of six hidden layers, each containing nine neurons, exhibited the lowest RMSE when evaluates on the testing dataset. The RSME is 38.37. Consequently, the investigation proceeded by utilizing the BP design consisting of six layers with 9 neurons.

In RF model, we systematically explore various hyperparameter values. We teste the 'tree' parameter with values ranging from 2 to 200,

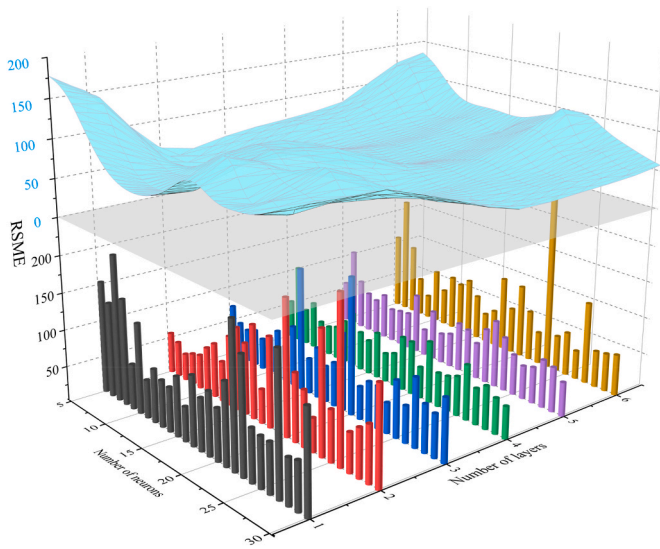


Fig. 11. Relationship between neural network layers and neurons.

finding that 45 trees yield the best performance. Similarly, the 'leaf node' parameter is evaluated with values from 1 to 10, with a minimum size of 1 demonstrating optimal results. The relevant results can be found in Table S1. In the LSTM algorithm, we test different hidden layers, ranging from 5 to 20, and ultimately select 12 as the optimal number of layers.

The residual error (RE) is also employed to evaluate the performance of ML models. RE is the difference between the y_i and y_i^* for the CHF predicted by the ML model. In simple terms, the residual represents the discrepancy between the predicted and actual values of the dependent variable in a regression model for a given observation. Residuals can be either positive or negative, depending on whether the observed value exceeds or falls short of the predicted value.

The set parameters could be seen at Table 4. Fig. 12 presents the results of ML models for training, testing, and RE at different samples. Comparing to other methods, the LSTM model exhibits high error derivation during the training period, with 75 data points showing RE values exceeding 100 and -100, separately. Similarly, the RF model also displays biases, where higher heating flux density corresponded to greater deviation. In contrast, the BP and SVR models demonstrate strong predictive performance in training. In the testing, the LSTM shows a noticeable error compared to other models. Within the -100 to 100 range, BP produced more predicted data with lower fluctuations.

Fig. 13 depicts the outcomes of evaluating inaccuracies for several ML algorithms. The R^2 values for the BP, RF, SVR, and LSTM models are 0.999, 0.992, 0.996, and 0.966, respectively, in the training set. The R^2

values obtained from the test data are 0.997 for BP, 0.991 for RF, 0.993 for SVR, and 0.962 for LSTM. In theory, the three ML models demonstrate excellent reliability, with the exception of the LSTM model. Both the training and testing data exhibit a significantly high RMSE over 100. The low prediction accuracy can be attributed to the limited generalization capability of the LSTM algorithm. Furthermore, the distribution of the CHF dataset also affects the outcome. The computation is inversely proportional to the number of layers: as the number of layers increases, the calculation time decreases. However, it is important to note that increasing the number of layers does not necessarily enhance the accuracy of the predicted outcomes.

LSTM networks involve numerous parameters (e.g., input gates, output gates, memory cells, etc.), typically encompassing multiple hidden layers and complex optimization procedures to calibrate these parameters. In cases of limited data volume, the need for capturing long-term time series becomes less apparent, and the training process may fail to identify the global optimal solution, resulting in suboptimal predictive performance. In contrast, conventional methods such as BP neural networks and SVRs are relatively simple and robust. They facilitate the identification of appropriate solutions through loss function optimization, allowing successful model training even with limited data.

Therefore, the LSTM algorithm is unsuitable for predicting CHF in narrow rectangular channels. The RF and SVR algorithms exhibit similar and consistent predictive performance, although the MPAE for the SVR model is slightly higher than that of the RF model. When analyzing the impact of the quantity of trees and leaf nodes, we observe a gradual decrease in computational speed. Among all methods, the BP approach exhibits outstanding performance, particularly in handling CHF data.

4.2. The BP neural network method optimization

Four distinct ML models are utilized to optimize the BP neural network. Table 5 displays the results for both training and testing. While the R^2 values testing remains consistent across all models, the GA-BP approach achieves the highest value of 0.999. In contrast, the PSO-BP and CS-BP methods exhibit higher error rates compared to other optimization algorithms and perform worse than the unoptimized BP neural network. Among the optimized models, the GA-BP method demonstrate superior performance, achieving an RMSE of 35.33 for both training and testing datasets.

Fig. 14 displays the distribution density of the RE for several models. During the training period, both the ACA-BP and GA-BP algorithms exhibit a consistent bell-shaped distribution, while the central axis of the PSO-BP and CS-BP algorithms is skewed towards the right. During the testing phase, the distribution for PSO-BP remains skewed to the right, whereas the ACA-BP algorithm shows an irregular distribution pattern. The GA is an optimization technique inspired by genetic evolution, employing mechanisms such as selection, crossover, and mutation during the optimization process. It effectively identifies optimal initial parameters and, through multi-generational evolution, determines the most suitable BP parameters for small data samples. This approach more effectively avoids local optima while improving the convergence speed and accuracy of the BP network. In contrast, due to their update mechanisms, PSO and CS are prone to converge on local optima, often exhibiting inconsistent performance in sparse data scenarios. This tendency may hinder the identification of the optimal parameter configuration for the BP network, resulting in diminished search efficiency.

Furthermore, we evaluate the RE performance of several ML models under varying levels of pressure, as depicted in Fig. 15. Due to the several predicted data points at various pressures, we utilize the average RE to assess the accuracy of the results. The average RE for CS-BP is notably higher at low pressure levels, exceeding 70. Conversely, the RE for the PSO-BP algorithm is significantly elevated under high pressure conditions, approaching a value of 70. In contrast, the GA-BP model demonstrates commendable performance across all pressure levels, exhibiting minimal RE and achieving the highest prediction accuracy.

Table 4

The optimized hyper-parameters in models established in the study.

ML models	Hyperparameter	Select value
BP neural network	Hidden layer	6
	Neurons	9
	Max iteration epoch	1000
	learning rate	1×10^{-6}
RF	Learning method	Levenberg-Marquardt
	The number of the tree	45
SVR	The number of the leaf	12
	Penalty factor	30000
LSTM	Gamma	0.01
	Max epochs	2000
	Initial learn rate	0.01
	Learn rate drop factor	0.1
	Learn rate drop period	1500
	Hidden layer	16

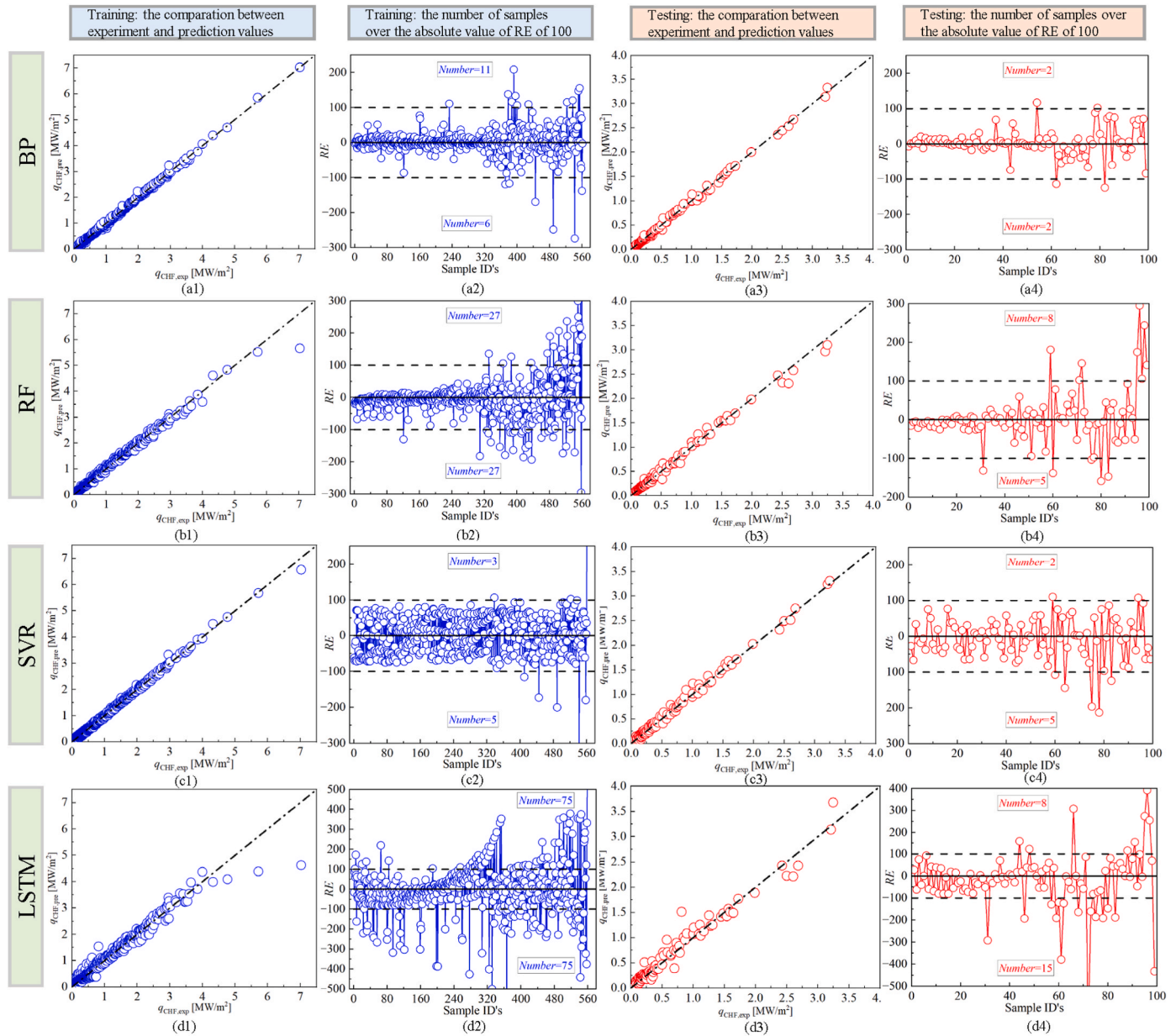


Fig. 12. The training, testing and RE for different ML methods (a) BP neural network, (b) RF model, (c) SVR, (d) LSTM.

5. Conclusion

In this study, we collected the CHF experimental data for narrow rectangular channel from recent years. Several CHF correlations are used for evaluation, and multiple ML models are employed to predict CHF values, with the models being optimized. The key findings can be summarized as follows.

- (1) The CHF dataset comprises a total of 660 experimental data points, encompassing a broad range of pressure values. Three methods for Pearson, Spearman, and Kendall are used for assessing the distribution of data. After examining the parameters of CHF, four input parameters are ultimately selected as inputs for training the ML models.
- (2) The existing CHF correlations are unsuitable for accurately predicting current data due to significant errors.
- (3) The overall performance of the ML models in CHF prediction is ranked as BP > RF > SVR > LSTM. The BP neural network,

optimized with 6 hidden layers (each containing 9 neurons), achieves accurate predictions, with a MAE of 24.33, a MPAAE of 0.049 and a RMSE of 38.37 for the test data.

- (4) Various ML strategies are used to enhance the performance of the BP neural network. The optimized approach for R^2 values consistently achieves values above 0.996 in both training and testing sections. Among the optimized models, the GA-BP neural network demonstrate the highest level of prediction accuracy, with a MAE of 23.80, a MPAAE of 0.0462, and a RMSE of 33.53. Additionally, the GA-BP approach accurately predicts CHF values across different pressure conditions.

The findings of this study indicate that ML techniques have the potential to serve as valuable models for the precise prediction of CHF data for narrow rectangular channels. However, certain models, such as the LSTM algorithm, may not be appropriate for training purposes. Additionally, some optimization techniques applied to the BP neural network may result in inferior prediction accuracy compared to the standard BP

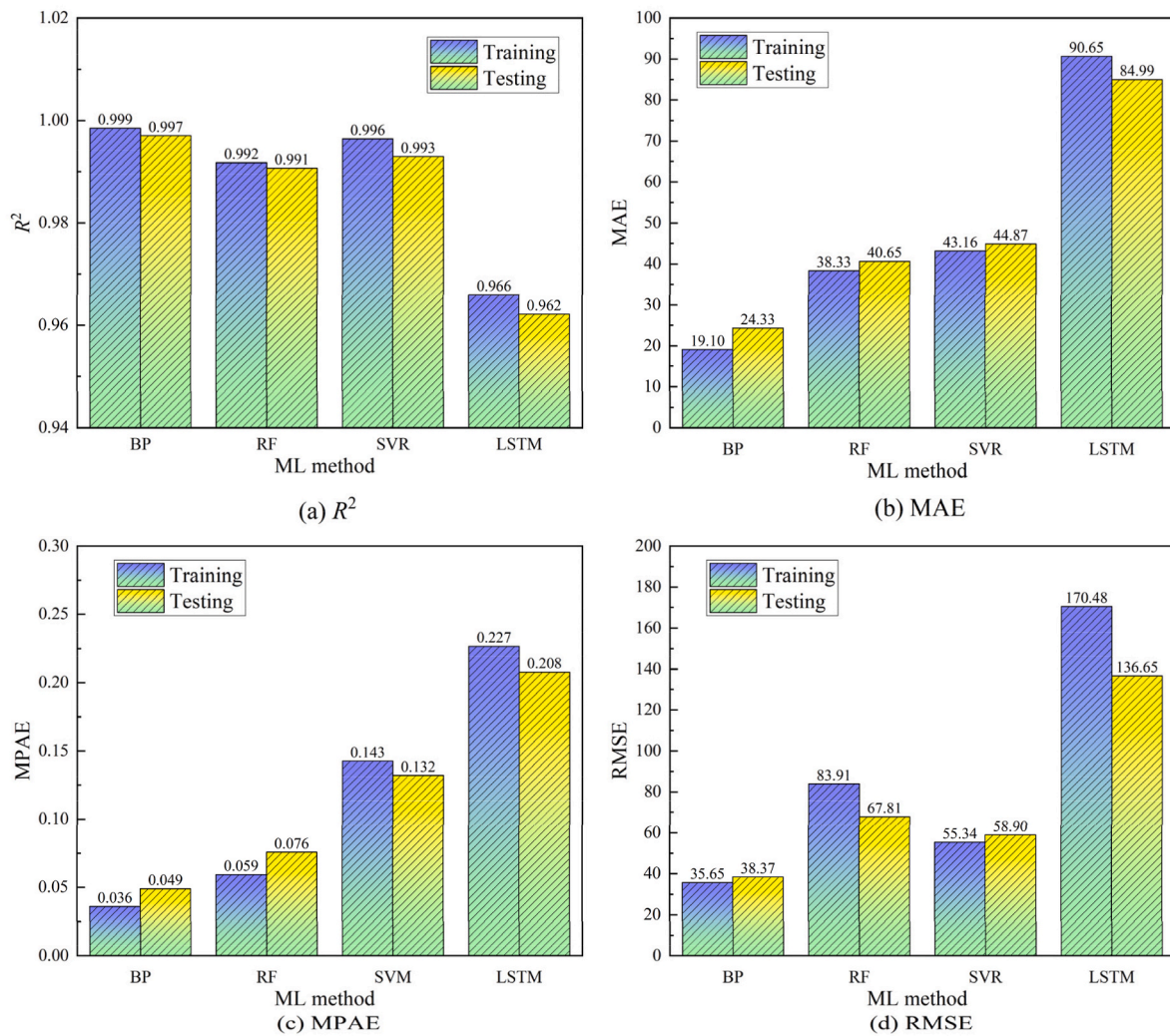


Fig. 13. The metrics for training and testing based different ML models.

Table 5
Comparative Evaluation of BP neural network for different optimization.

Metrics	PSO-BP		ACA-BP		GA-BP		CS-BP	
	Training	Testing	Training	Testing	Training	Testing	Training	Testing
R^2	0.997	0.996	0.998	0.997	0.999	0.997	0.998	0.997
MAE	31.693	31.752	20.730	25.581	19.132	23.799	23.077	26.117
MPAA	0.0835	0.0798	0.0559	0.0569	0.0462	0.0581	0.0501	0.0637
RMSE	51.43	45.33	37.98	39.96	33.53	37.65	43.65	40.56

approach. Hence, it is crucial to train ML models using appropriate techniques. If the training dataset is insufficient, the BP neural network is prone to memorizing every detail of the training set, leading to significantly poor performance on the test set. In such cases, the BP model may achieve perfect fit on the training data but fail to generalize to new data. When the GA is used to optimize the BP neural network, GA can assist in selecting an appropriate set of parameters during the search process. However, overfitting remains a potential issue when the training dataset is limited. Although GA mitigates early overfitting, if the dataset is extremely small, GA may still overfit the existing data through its evolutionary processes. Future research should integrate a broader spectrum of operational parameters, including geometric and thermal-hydraulic factors, as well as a diverse range of applications. To enhance model accuracy, it is crucial to acquire a more extensive dataset of CHF data that covers a wider variety of operational conditions. To

further enhance model accuracy, it is necessary to acquire a larger dataset of CHF data covering a broader range of operational conditions. Based on the available CHF data points, testing additional ML algorithms is recommended to identify the most effective approach for prediction, thereby ensuring more reliable and robust outcomes.

CRediT authorship contribution statement

Enpei Wang: Writing – review & editing, Writing – original draft, Supervision, Software, Resources, Project administration, Methodology, Conceptualization. **Meng Zhao:** Resources, Project administration, Methodology. **Haopeng Shi:** Software, Resources, Investigation, Funding acquisition. **Hao Zhang:** Software, Investigation, Funding acquisition, Formal analysis. **Yanhua Yang:** Resources, Project administration, Methodology, Investigation. **Qinglong Wen:** Software, Resources,

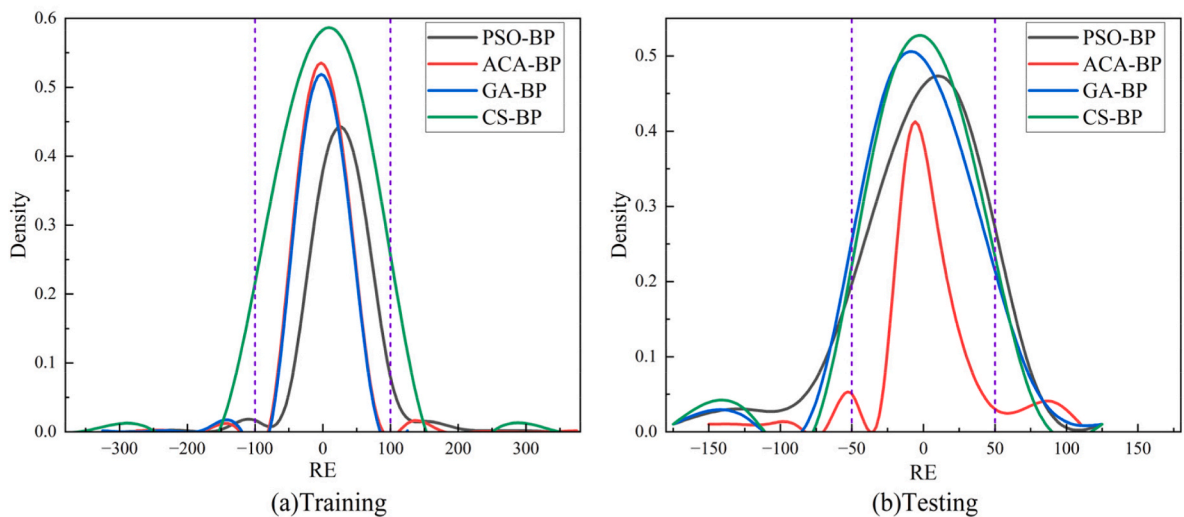


Fig. 14. The RE performance for training and testing based on different ML models.

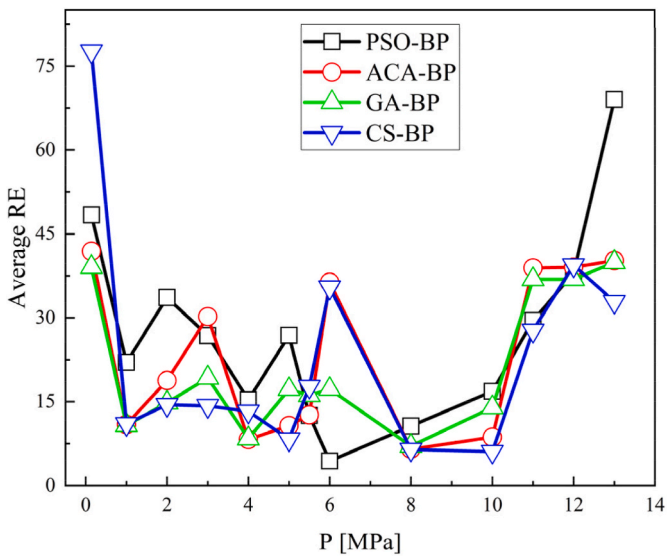


Fig. 15. The distribution for average RE with pressure increasing based different algorithms.

Project administration, Methodology.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ijthermalsci.2025.110050>.

Nomenclature	
A	Area, [m ²]
D_{hed}	Hydraulic equivalent diameter, [mm]
H	Rectangular channel height, [mm]
H_{BO}	Enthalpy at burnout location, [J/kg]
G	Mass flux, [kg/m ² s]
g	Acceleration of gravity, [m/s ²]
L	Length, [m]
P	Pressure, [Pa]
q_{CHF}	Critical heat flux, [kW/m ²]
q_{CHF}^*	Dimensionless critical heat flux [–]
ΔT_{in}	Inlet subcooling, [K]

(continued on next page)

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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(continued)

Nomenclature	
U	Velocity, [m/s]
W	Rectangular channel width, [mm]
λ	Critical wavelength
Abbreviation	
ACA	Ant colony algorithm
ANN	Artificial neural network
BP	Back-propagation
CHF	Critical heat flux
CS	Cuckoo search
GA	Genetic algorithms
LSTM	Long short-term memory
ML	Machine learning
PSO	Particle swarm optimization
RF	Random forest
SVM	Support vector machine
SVR	Support vector regression

Data availability

Data will be made available on request.

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